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Screenshots

Console

Terminal

Background Jobs

R • R 4.6.1 • ~/Documents/R Programming/NPTEL/

> list1 = list(1,2,3)

> list2 = list("water", "juice", "lemonade")

> list12 = c(list1, list2)

> list12

[[1]]

[1] 1

[[2]]

[1] 2

[[3]]

[1] 3

[[4]]

[1] "water"

[[5]]

[1] "juice"

[[6]]

[1] "lemonade"

> unlist(list1)

[1] 1 2 3

> unlist(list2)

[1] "water" "juice" "lemonade"

> mode(list1)

[1] "list"

> mode(unlist(list1))

[1] "numeric"

> append(list1, 100)

[[1]]

[1] 1

[[2]]

[1] 2

[[3]]

[1] 3

[[4]]

[1] 100

	Console	Terminal	Background Jobs
	R R 4.6.1 · ~/Documents/R Programming/NPTEL/ <pre>> append(list2, "coffee") [[1]] [1] "water" [[2]] [1] "juice" [[3]] [1] "lemonade" [[4]] [1] "coffee" > append(list1, 100, after = 2) [[1]] [1] 1 [[2]] [1] 2 [[3]] [1] 100 [[4]] [1] 3 > append(list2, "coffee", after = 2) [[1]] [1] "water" [[2]] [1] "juice" [[3]] [1] "coffee" [[4]] [1] "lemonade"</pre>		

	Console	Terminal	Background Jobs
	R 4.6.1 · ~/Documents/R Programming/NPTEL/ <pre>> list1 = list(1,2,3,4,5,6) > list2 = list("water", "juice", "lemonade", "tea", "coffee", "milk") > list1[-2] [[1]] [1] 1 [[2]] [1] 3 [[3]] [1] 4 [[4]] [1] 5 [[5]] [1] 6 > list2[-1] [[1]] [1] "juice" [[2]] [1] "lemonade" [[3]] [1] "tea" [[4]] [1] "coffee" [[5]] [1] "milk" > list1[2:4] [[1]] [1] 2 [[2]] [1] 3 [[3]] [1] 4</pre>		

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Console Terminal Background Jobs
R - R 4.6.1 - ~/Documents/R Programming/NPTEL/

> list1[c(1,3,5)]
[[1]]
[1] 1

[[2]]
[1] 3

[[3]]
[1] 5

> list2[2:4]
[[1]]
[1] "juice"

[[2]]
[1] "lemonade"

[[3]]
[1] "tea"

> list2[c(1,3,5)]
[[1]]
[1] "water"

[[2]]
[1] "lemonade"

[[3]]
[1] "coffee"

> x = 1:10
> x[(x > 5)]
[1] 6 7 8 9 10
> x[(x %% 2 == 0)]
[1] 2 4 6 8 10
> x[(x %% 2 == 1)]
[1] 1 3 5 7 9
> x[5] = NA
> y = x[!is.na(x)]
> mean(x)
[1] NA
> mean(y)
[1] 5.555556
> x = 1:10
> x[-(1:5)]
[1] 6 7 8 9 10
```

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Console Terminal Background Jobs
R - R 4.6.1 - ~/Documents/R Programming/NPTEL/

> x[6:10]
[1] 6 7 8 9 10
> z = list(a1 = 1, a2 = "c", a3 = 1:3)
> z
$a1
[1] 1

$a2
[1] "c"

$a3
[1] 1 2 3

> names(z)
[1] "a1" "a2" "a3"
> names(z)[3] = "c2"
> z
$a1
[1] 1

$a2
[1] "c"

$c2
[1] 1 2 3

> x = c(water=1, juice=2, lemonade=3)
> names(x)
[1] "water" "juice" "lemonade"
> x["juice"]
juice
2
> x = 1:10
> x[]
[1] 1 2 3 4 5 6 7 8 9 10
> ab = list(1, 2, 3, "X", "Y", "Z")
> dim(ab) = c(2,3)
> print(ab)
      [,1] [,2] [,3]
[1,] 1     3     "Y"
[2,] 2     "X"   "Z"
> mode(print(ab))
      [,1] [,2] [,3]
[1,] 1     3     "Y"
[2,] 2     "X"   "Z"
[1] "list"
```

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Console Terminal Background Jobs
R - R 4.6.1 - ~/Documents/R Programming/NPTEL/
> x = c(3, 4, 5, 6, 1, 2, 3, 3, 4, 4, 5, 6)
> y = as.factor(x)
> y
[1] 3 4 5 6 1 2 3 3 4 4 5 6
Levels: 1 2 3 4 5 6
> x = factor(c("lemonade", "lemonade", "juice", "lemonade", "water"))
> x
[1] lemonade lemonade juice   lemonade water
Levels: juice lemonade water
> help("factor")
> y = c(1, 4, 3, 5, 4, 2, 4)
> possible.dieface = c(1, 2, 3, 4, 5, 6)
> labels.dieface = c("one", "two", "three", "four", "five", "six")
> facy = factor(y, levels = possible.dieface, labels = labels.dieface)
> facy
[1] one  four  three five  four  two   four
Levels: one two three four five six
> x = c(3, 4, 5, 6, 1, 2, 3, 3, 4, 4, 5, 6)
> y = as.factor(x)
> y
[1] 3 4 5 6 1 2 3 3 4 4 5 6
Levels: 1 2 3 4 5 6
>
> x = factor(c("lemonade", "lemonade", "juice", "lemonade", "water"))
> x
[1] lemonade lemonade juice   lemonade water
Levels: juice lemonade water
> class(9)
[1] "numeric"
> class("9")
[1] "character"
> class(print)
[1] "function"
> x = matrix(nrow=2, ncol=2, data=1:4)
> class(x)
[1] "matrix" "array"
> brands = c("A", "A", "B", "B", "B", "B", "C")
> brands_fac = factor(brands)
> brands_fac
[1] A A B B B B C
Levels: A B C
> unclass(brands_fac)
[1] 1 1 2 2 2 2 3
attr(,"levels")
[1] "A" "B" "C"

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Console Terminal Background Jobs
R - R 4.6.1 - ~/Documents/R Programming/NPTEL/
> colours = c("blue", "green", "red")
> colours[unclass(brands_fac)]
[1] "blue" "blue" "green" "green" "green" "green" "red"
> x = factor(c("lemonade", "lemonade", "juice", "lemonade", "water"))
> unclass(x)
[1] 2 2 1 2 3
attr(,"levels")
[1] "juice" "lemonade" "water"
> x = factor(c("lemonade", "lemonade", "juice", "lemonade", "water"),
+           levels=c("water", "juice", "lemonade"))
> x
[1] lemonade lemonade juice  lemonade water
Levels: water juice lemonade
> unclass(x)
[1] 3 3 2 3 1
attr(,"levels")
[1] "water" "juice" "lemonade"
> levels(x)
[1] "water" "juice" "lemonade"
> income = ordered(c("high", "high", "low", "medium", "medium"),
+                 levels=c("low", "medium", "high"))
> income
[1] high  high  low   medium medium
Levels: low < medium < high
> unclass(income)
[1] 3 3 1 2 2
attr(,"levels")
[1] "low" "medium" "high"
> print(sqrt(2))
[1] 1.414214
> print(sqrt(2), digits=5)
[1] 1.4142
> print(sqrt(2), digits=10)
[1] 1.414213562
> print("apple")
[1] "apple"
> print(c("apple", "banana"))
[1] "apple" "banana"
> print(c("apple", "banana", 6, 10))
[1] "apple" "banana" "6" "10"
> format(0.5)
[1] "0.5"
> print(format(0.5, digits=10, nsmall=15))
[1] "0.500000000000000"
> format(c("A", "BB", "CCC", "DDDD"), width=7, justify="centre")
[1] " A " " BB " " CCC " " DDDD "
> format(c("A", "BB", "CCC", "DDDD"), width=7, justify="centre")
[1] " A " " BB " " CCC " " DDDD "
> format(c("A", "BB", "CCC", "DDDD"), width=7, justify="left")
[1] "A " " BB " " CCC " " DDDD "
> format(c("A", "BB", "CCC", "DDDD"), width=7, justify="right")
[1] " A " " BB " " CCC " " DDDD "
> format(c("A", "BB", "CCC", "DDDD"), width=7, justify="none")
[1] "A" "BB" "CCC" "DDDD"
> x = matrix(nrow=3, ncol=2, data=1:6, byrow=T)
> print(x)
      [,1] [,2]
[1,] 1 2
[2,] 3 4
[3,] 5 6
> format(1234567, big.mark=",")
[1] "1,234,567"
> format(12345678, big.mark=",")
[1] "12,345,678"
> format(123456789, big.mark=",")
[1] "123,456,789"
> format(123456789, big.mark=" ")
[1] "123 456 789"
>

```

R ▾ Global Environment ▾		
Data		
ab	List of 6	
list1	List of 6	
list12	List of 6	
list2	List of 6	
x	int [1:3, 1:2] 1 3 5 2 4 6	
z	List of 3	
Values		
brands	chr [1:7] "A" "A" "B" "B" "B" "B" "C"	
brands_fac	Factor w/ 3 levels "A","B","C": 1 1 2 2 2 2 3	
colours	chr [1:3] "blue" "green" "red"	
facy	Factor w/ 6 levels "one","two","three",...: 1 4 3 5 4 2 4	
income	Ord.factor w/ 3 levels "low"<"medium"<...: 3 3 1 2 2	
labels.dieface	chr [1:6] "one" "two" "three" "four" "five" "six"	
possible.dieface	num [1:6] 1 2 3 4 5 6	
y	labels.dieface (character , 432 bytes) s "1","2","3","4",...: 3 4 5 6 1 2 3 3 4 4 ...	